

AMD DEVELOPER HACKATHON: ACT II · VIDEO CAPTIONING TRACK

StyleCap

One video in. **Four voices out.**

Formal

Sarcastic

Humorous · tech

Humorous · everyday

01 The task

Track 2 is harness-judged: the container receives video URLs and must return one caption per requested style, faithful to the footage and true to the tone. An LLM judge scores every caption on **accuracy** and **style match**.

/input/tasks.json

```
{ "task_id": "1",  
  "video_url": "https://...clip.mp4",  
  "styles": ["formal", "sarcastic",  
            "humorous_tech", "humorous_non_tech"] }
```

/output/results.json

```
{ "task_id": "1",  
  "captions": {  
    "formal": "...", "sarcastic": "...",  
    "humorous_tech": "...", ... } }
```

Hidden evaluation set: ~12 unseen clips, 30s to 2min, spanning nature, urban, animals, people, sports, food, weather, and technology.

02 How it works

STEP 1

Download

Fetch each clip; 4 tasks run in parallel workers.

STEP 2

Sample frames

8 evenly spaced frames via OpenCV, in memory, downscaled to 768px.

STEP 3

One VLM call

A single multimodal request to Fireworks AI (kimi-k2p6) returns all four styles at once.

STEP 4

Validate & write

Robust JSON parsing, then exact harness-format output.

Tone quality is engineered, not hoped for

Few-shot tone exemplars are baked into the prompt. They were selected with our own LLM-judge evaluation harness (`scripts/eval_captions.py`), which scores candidate prompts on the same two dimensions the judges use.

03 Built for the harness

~20s

3 official UHD clips end to end (10 min budget)

1 call

per video, all four styles at once

1.21 GB

linux/amd64 image (10 GB limit)

- Exit code 0, valid JSON, every requested style always present
- Public image: `ghcr.io/ov3rkill15/amd-video-captioning:latest`, anonymously pullable
- Container starts instantly (no model weights inside; inference is serverless via API)

04 A broken clip never sinks the run

Three fallback layers guarantee a complete, valid answer for every task:

LAYER 1

API retries

Transient errors and rate limits are retried with backoff.

LAYER 2

Per-style recovery

If the combined call fails to parse, each style is requested individually.

LAYER 3

Guaranteed captions

As a last resort, safe generic captions are emitted, so no style is ever missing.

Missing styles score zero for a clip. With this design, that failure mode is structurally impossible: the container always exits 0 with a complete `results.json`.

05 Sample output · official kitten clip

FORMAL

A small orange kitten walks slowly toward the camera through a shaded path bordered by dense green foliage and tree trunks.

SARCASTIC

Breaking news: a kitten walks forward. Nature documentaries will never recover from this level of drama.

HUMOROUS · TECH

Kernel panic averted. This orange fluff successfully compiled a walk.exe with zero segmentation faults.

HUMOROUS · EVERYDAY

This kitten is walking toward you like it already knows you have no treats and does not care.

06 Generalisation, not memorisation

- Prompts are purely style-driven; nothing is tuned to the example clips
- Same pipeline handles nature, urban, animals, people, sports, food, weather, tech footage
- Provider-agnostic client (OpenAI-compatible), model swappable via env config
- Modular repo: `src/pipeline` · `src/models` · `scripts/eval_captions.py` · local web demo UI

Image

`ghcr.io/ov3rkill15/amd-video-captioning:latest`

Source

`github.com/0v3rkill15/amd-video-captioning`